

Onyx — AudioKit reference

Onyx · multi-process OS on Raspberry Pi 4

The reference of **AudioKit** (`SD:/Lib/audiokit.so`, `user/Kits/audiokit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: [The Kits](#).

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What it is

AudioKit is sound: files played, notes on the General MIDI instruments, the FM synthesizer, mixing, WAV files written, and the system's output for a program that makes its own samples.

Include	<code>#include "audiokit/audiokit.h"</code>
Link	<code>lib/audiokit.imp.a</code>
Library	<code>SD:/lib/audiokit.so</code> — 232 entries in its table (<code>user/Kits/audiokit/audiokit.abi</code> , append-only)
Sources	<code>user/Kits/audiokit/</code>

Using it

Several programs play at once: the system mixes them, each with its own volume (Control Panel ► Sound).

A sound file (WAV, MP3, OGG, FLAC, MIDI, FM Tracker songs), played in the background:

```
#include "audiokit/audiokit.h"
```

```
if (ak_play ("SD:/music/song.mp3", 0) < 0) ax_putln (ak_play_error ());
ak_play_volume (80);                                     // 0..100
/* ... the program goes on ... */
ak_play_stop ();
```

Notes (the General MIDI instruments, a SoundFont):

```
ak_note_on (0, 60, 100);                                // channel, key (60 = middle C),
velocity
kapi_msleep (400);
ak_note_off (0, 60);
```

A sound effect (the FM synthesizer's voices — what the games use):

```
ak_fm_start (0, 440000, 0, 200);           // voice 0, 440 Hz (in milli-Hz),  
volume 0..255  
kapi_msleep (120);  
ak_fm_stop (0);
```

The program's own samples (an emulator, a synthesizer): stereo, 16 bits, 44.1 kHz (AUDIOKIT_RATE).

```
if (ak_out_open (256, 1024) == 1)  
{  
    short frames[256 * 2];  
    /* fill frames ... */  
    ak_out_write (frames, 256);           // waits for room  
    ak_out_close ();  
}
```

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Everything the headers declare, in their order — the details are in each header's part below.

Name	What it does	Header
ak_info	(a type)	audiokit.h
ak_open	The file opened (its kind from its extension, then its first bytes) -> 0	audiokit.h
ak_read	Up to frames frames at AUDIOKIT_RATE into out (2 shorts each) -> how many (0	audiokit.h
ak_seek_ms	1 done / 0	audiokit.h
ak_info_of		audiokit.h
ak_close		audiokit.h
ak_tags	What a sound file says about itself (MP3	audiokit.h
ak_tags_read	1 / 0: not a sound file of ours, unreadable	audiokit.h
ak_play	0 started / -1 (ak_play_error says why)	audiokit.h
ak_play_stop		audiokit.h
ak_play_pause		audiokit.h
ak_play_state	AK_* (the file's; the live notes aside)	audiokit.h
ak_play_pos_ms		audiokit.h
ak_play_len_ms		audiokit.h
ak_play_seek_ms		audiokit.h
ak_play_volume	0..100 (-1: only ask) -> the	audiokit.h

Name	What it does	Header
	volume	
<code>ak_play_error</code>	the last error's words ("": none)	<code>audiokit.h</code>
<code>ak_play_wait</code>	waits while a file plays, ms at most (-1: no limit) -> its state	<code>audiokit.h</code>
<code>ak_play_keep_output</code>	on	<code>audiokit.h</code>
<code>ak_note_on</code>	0 / -1 (no SoundFont: <code>ak_play_error</code>)	<code>audiokit.h</code>
<code>ak_note_off</code>	Live notes on the General MIDI synthesizer (the default SoundFont), 16 channels (9)	<code>audiokit.h</code>
<code>ak_program</code>	the channel's instrument, 0..127 (General MIDI)	<code>audiokit.h</code>
<code>ak_control</code>	a MIDI controller (7 volume, 10 pan, 64 pedal...)	<code>audiokit.h</code>
<code>ak_pitch_bend</code>	-8192..8191	<code>audiokit.h</code>
<code>ak_notes_off</code>	everything silent now	<code>audiokit.h</code>
<code>ak_out_open</code>	<code>kapi_sound_config</code> 's -> 1 ours, 0 busy, -1 no sound	<code>audiokit.h</code>
<code>ak_out_write</code>	all of them, waiting for room -> n (< n: closed)	<code>audiokit.h</code>
<code>ak_out_free</code>	frames that would be taken without waiting	<code>audiokit.h</code>
<code>ak_out_queued</code>	frames written and not yet played	<code>audiokit.h</code>
<code>ak_out_close</code>	(Not together with the player)	<code>audiokit.h</code>
<code>ak_gain_s16</code>		<code>audiokit.h</code>
<code>ak_mix_s16</code>	<code>dst += src * gain</code> , saturated	<code>audiokit.h</code>
<code>ak_mono_to_stereo</code>	the first <code>frames</code> shorts spread, in place	<code>audiokit.h</code>
<code>ak_volume_gain</code>	0..100 on the ear's curve -> a 16.16 gain	<code>audiokit.h</code>
<code>ak_resampler_new</code>		<code>audiokit.h</code>
<code>ak_resample</code>	<code>in_frames</code> frames in -> the frames written to out (<code>out_cap</code> at most)	<code>audiokit.h</code>

Name	What it does	Header
<code>ak_resampler_free</code>		<code>audiokit.h</code>
<code>ak_f32_to_s16</code>	(FPU programs) Two float channels (-1..1) -> 16-bit stereo, through the soft limiter (Koton's	<code>audiokit.h</code>
<code>ak_soft_clip</code>		<code>audiokit.h</code>
<code>ak_note_mhz</code>	a MIDI key's frequency, milli-Hz (69 = A4 = 440000)	<code>audiokit.h</code>
<code>ak_note_key</code>	a note (0 = C .. 11 = B) and an octave -> the key (C4 = 60)	<code>audiokit.h</code>
<code>ak_note_octave_mhz</code>	... its frequency, milli-Hz	<code>audiokit.h</code>
<code>ak_note_name</code>	"C4", "F#3" (60 = C4)	<code>audiokit.h</code>
<code>ak_note_parse</code>	"C4", "f#3", "Bb2" -> the key, -1	<code>audiokit.h</code>
<code>ak_wav_header</code>	The 44-byte header of a 16-bit PCM file of <code>data_bytes</code> bytes of samples -> 44.	<code>audiokit.h</code>
<code>ak_wav_save</code>	stereo -> 0 / -1	<code>audiokit.h</code>
<code>ak_wav_begin</code>	0: not created	<code>audiokit.h</code>
<code>ak_wav_write</code>	-> the frames taken	<code>audiokit.h</code>
<code>ak_wav_end</code>	0 / -1	<code>audiokit.h</code>
<code>ak_soundfont_default</code>	The default SoundFont	<code>audiokit.h</code>
<code>ak_soundfont_name</code>	its name ("": none loaded)	<code>audiokit.h</code>
<code>ak_soundfont_find</code>	A SoundFont's file	<code>audiokit.h</code>
<code>ak_soundfont_load</code>	-> an <code>ms::SoundFont *</code> of the caller's (0: err)	<code>audiokit.h</code>
<code>ak_soundfont_free</code>		<code>audiokit.h</code>
<code>ak_soundfont_prefer</code>	The file the default SoundFont is to be (a program's setting	<code>audiokit.h</code>
<code>ak_synth_new</code>	on the default SoundFont, at <code>AUDIOKIT_RATE</code> -> 0: none	<code>audiokit.h</code>
<code>ak_synth_free</code>		<code>audiokit.h</code>
<code>ak_synth_midi</code>	0x90 note on, 0x80 off, 0xC0 program...	<code>audiokit.h</code>
<code>ak_synth_render</code>	never allocates, never	<code>audiokit.h</code>

Name	What it does	Header
	calls the kernel	
ak_fm_instrument	0 / -1	audiokit.h
ak_fm_start	voice 0..15, volume 0..255	audiokit.h
ak_fm_stop	-1: all (the releases are heard)	audiokit.h
ak_fm_silence	all silent at once	audiokit.h
ak_fm_render	the voices' next frames written to out (off line: an export)	audiokit.h
ak_fm_live	0: the player leaves the voices alone -- the program renders them (ak_fm_render); 1 (at first): heard	audiokit.h
ak_reverb_new		audiokit.h
ak_reverb_set	< 0: unchanged	audiokit.h
ak_reverb_process	mono in, stereo out	audiokit.h
ak_reverb_mute		audiokit.h
ak_reverb_free		audiokit.h
ak_chorus_new	(MellySynth's own: 0.002, 0.0019, 0.4)	audiokit.h
ak_chorus_process		audiokit.h
ak_chorus_mute		audiokit.h
ak_chorus_free		audiokit.h

audiokit/audiokit.h

audiokit.h -- AudioKit, Onyx's shared sound library (SD:/lib/audiokit.so; docs/03 "AudioKit").

Everything about sound that more than one program can use, in one copy for the whole system:

```

files      a sound file of any kind -- MP3, FLAC, WAV, Ogg Vorbis, MIDI (through a
SoundFont) --
            read as 16-bit stereo frames at the output's rate (ak_open, ak_read, ak_seek_ms)
the player a file played in the background on the system's output, and notes played live on
a
            General MIDI synthesizer, mixed (ak_play, ak_note_on ...): one line to make a
sound
the output the system's sound output for a program that makes its own frames (ak_out_*)
mixing      gain, mix with saturation, mono to stereo, float to 16 bits with a soft limiter,
            a rate converter (ak_mix_s16, ak_f32_to_s16, ak_resample_*)
notes       a note's frequency, its name, a name's note (ak_note_*)
WAV         a WAV file written (ak_wav_header, ak_wav_save)
the synthesizer MeltySynth, the SoundFont synthesizer of Koton: its C++ interface
            (Apps/koton/synth/meltysynth.h: ms::Synthesizer, ms::soundfont_load, the reverb
and

```

```

        the chorus) is exported as it is; ak_synth_* is the same for C and for BASIC
the decoders      minimp3, dr_flac, dr_wav and stb_vorbis: their own C interfaces
        (Apps/media/codecs.h) are exported as they are, for a program that wants them

```

raw

A program links lib/audiokit.imp.a (user/Makefile) and calls these as plain functions: the library is opened before main (user/Runtime/lib.h). The interface is append-only (audiokit/audiokit.abi). Frames are always interleaved 16-bit stereo (left, right) unless said; "the rate" is AUDIOKIT_RATE, the system output's (44100 Hz).

The entries that take or return float / double are for programs built with the FPU (every newlib app, /bin/basic, the games built with CXXFLAGS_FP): an integer-only program (-mgeneral-regs-only) uses the others -- all the file, player, output, note and 16-bit mixing calls are integer.

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```

#define AUDIOKIT_RATE    44100           // the frames of every call: 16-bit stereo at this
rate

```

a sound file

```

struct ak_info
{
    int rate;           // the file's own rate (ak_read gives AUDIOKIT_RATE)
    int channels;       // what the file has (ak_read gives stereo)
    int bits;          // its sample size (0: not known)
    int kbps;           // its bit rate (0: not known)
    long long length_ms; // its length (0: not known)
    char format[8];     // "MP3", "FLAC", "WAV", "OGG", "MIDI", "FMS"
};
typedef struct ak_stream ak_stream;

```

The file opened (its kind from its extension, then its first bytes) -> 0: err (cap bytes) says why. A MIDI file plays through the default SoundFont (ak_soundfont_default); an FM Song (.fms, FM Tracker's) on the FM synthesizer (one read at a time per process).

```

ak_stream *ak_open (const char *path, char *err, int cap);

```

Up to `frames` frames at `AUDIOKIT_RATE` into out (2 shorts each) -> how many (0: the end).

```
int ak_read (ak_stream *s, short *out, int frames);
int ak_seek_ms (ak_stream *s, long long ms);           // 1 done / 0
void ak_info_of (ak_stream *s, struct ak_info *out);
void ak_close (ak_stream *s);
```

What a sound file says about itself (MP3: ID3v2 / ID3v1; FLAC and Ogg: Vorbis comments; WAV: LIST INFO; MIDI: the first track's name; an FM Song: its title and author) -- read only, nothing decoded. What is missing comes from the path: the file's name (a leading "03 - ": the track), its folder (the album), the folder above (the artist).

```
struct ak_tags
{
    char title[128], artist[96], album_artist[96], album[128], genre[48];
    int year, track, disc;           // 0: not known
    int duration_ms;                 // 0: not known
    char format[8];                  // "MP3", "OGG", "FLAC", "WAV", "MIDI", "FMS"
    long long cover_offset;           // a picture inside the file (JPEG / PNG bytes): where,
    and
    unsigned cover_length;           // how long (0: none)
    int reserved[8];
};
int ak_tags_read (const char *path, struct ak_tags *out); // 1 / 0: not a sound file of
ours, unreadable
```

the player: a file in the background, notes played live

One per process: a thread of its own that holds the system's output while it has something to play (another program playing: `AK_BUSY` until it lets go). The file and the live notes are mixed.

```
#define AK_STOPPED 0
#define AK_PLAYING 1
#define AK_PAUSED 2
#define AK_BUSY 3           // playing, but another program holds the output

int ak_play (const char *path, int loop);           // 0 started / -1 (ak_play_error says why)
void ak_play_stop (void);
void ak_play_pause (int on);
int ak_play_state (void);           // AK_* (the file's; the live notes aside)
long long ak_play_pos_ms (void);
long long ak_play_len_ms (void);
int ak_play_seek_ms (long long ms);
int ak_play_volume (int volume);           // 0..100 (-1: only ask) -> the volume
const char *ak_play_error (void);           // the last error's words ("": none)
int ak_play_wait (int ms);           // waits while a file plays, ms at most (-1: no
limit) -> its state
```

on: the player keeps the system's output once it has it, also when it has nothing to play (a program that plays the kernel's voices as well: BASIC's `SOUND` / `PLAY`) -- else it lets go after ~0.6 s of silence, so that other programs can play.

```
void ak_play_keep_output (int on);
```

Live notes on the General MIDI synthesizer (the default SoundFont), 16 channels (9: the drums).

```
int ak_note_on (int channel, int key, int velocity); // 0 / -1 (no SoundFont:
ak_play_error)
```

```

void ak_note_off (int channel, int key);
void ak_program (int channel, int program);    // the channel's instrument, 0..127 (General
MIDI)
void ak_control (int channel, int controller, int value);    // a MIDI controller (7 volume,
10 pan, 64 pedal...)
void ak_pitch_bend (int channel, int value);    // -8192..8191
void ak_notes_off (void);    // everything silent now

```

the system's output, for a program that makes its own frames

(Not together with the player: both want the output.)

```

int ak_out_open (int chunk_frames, int ahead);    // kapi_sound_config's -> 1 ours, 0 busy,
-1 no sound
int ak_out_write (const short *frames, int n);    // all of them, waiting for room -> n (<
n: closed)
int ak_out_free (void);    // frames that would be taken without waiting
int ak_out_queued (void);    // frames written and not yet played
void ak_out_close (void);

```

mixing and conversion (16-bit stereo frames; gains in 16.16: 65536 = 1)

```

void ak_gain_s16 (short *buf, int frames, int gain);
void ak_mix_s16 (short *dst, const short *src, int frames, int gain);    // dst += src * gain,
saturated
void ak_mono_to_stereo (short *buf, int frames);    // the first `frames` shorts
spread, in place
int ak_volume_gain (int volume);    // 0..100 on the ear's curve -> a 16.16 gain

```

```

typedef struct ak_resampler ak_resampler;    // a rate converter (linear), stereo
ak_resampler *ak_resampler_new (int in_rate, int out_rate);

```

in_frames frames in -> the frames written to out (out_cap at most); *used: the input frames taken.

```

int ak_resample (ak_resampler *r, const short *in, int in_frames, short *out, int out_cap,
int *used);
void ak_resampler_free (ak_resampler *r);

```

(FPU programs) Two float channels (-1..1) -> 16-bit stereo, through the soft limiter (Koton's: straight up to 0.75, then a knee that never passes 1) instead of clipping.

```

void ak_f32_to_s16 (const float *left, const float *right, short *out, int frames, float
gain);
float ak_soft_clip (float x);

```

notes

```

int ak_note_mhz (int key);    // a MIDI key's frequency, milli-Hz (69 = A4 =
440000)
int ak_note_key (int note, int octave);    // a note (0 = C .. 11 = B) and an octave ->
the key (C4 = 60)
int ak_note_octave_mhz (int note, int octave);    // ... its frequency, milli-Hz
void ak_note_name (int key, char *out8);    // "C4", "F#3" (60 = C4)
int ak_note_parse (const char *name);    // "C4", "f#3", "Bb2" -> the key, -1

```

WAV

The 44-byte header of a 16-bit PCM file of data_bytes bytes of samples -> 44.

```
int ak_wav_header (unsigned char *out44, int rate, int channels, unsigned data_bytes);
int ak_wav_save (const char *path, const short *frames, int n, int rate); // stereo -> 0 / -1
```

A long file written as it is made (an export): its length first, then its frames, then the end.

```
typedef struct ak_wav ak_wav;
ak_wav *ak_wav_begin (const char *path, int rate, int channels, long long frames); // 0: not created
int ak_wav_write (ak_wav *w, const short *frames, int n); // -> the frames taken
int ak_wav_end (ak_wav *w); // 0 / -1
```

the synthesizer, for C and BASIC (C++: meltyynth.h's ms::Synthesizer is exported too)

The default SoundFont: the first .sf2 of SD:/res/soundfonts (the package GeneralUser GS), loaded once per process -> an ms::SoundFont *, 0: err says why.

```
void *ak_soundfont_default (char *err, int cap);
const char *ak_soundfont_name (void); // its name ("": none loaded)
```

A SoundFont's file: preferred if it exists, else the first .sf2 of SD:/res/soundfonts, SD:/koton/soundfonts, SD:/music/soundfonts, SD:/music, SD:/apps/koton.app -> 1 (out: its path) / 0.

```
int ak_soundfont_find (const char *preferred, char *out, int cap);
void *ak_soundfont_load (const char *path, char *err, int cap); // -> an ms::SoundFont * of the caller's (0: err)
void ak_soundfont_free (void *sf);
```

The file the default SoundFont is to be (a program's setting; before the first MIDI sound). "": none.

```
void ak_soundfont_prefer (const char *path);
typedef struct ak_synth ak_synth;
ak_synth *ak_synth_new (void); // on the default SoundFont, at AUDIOKIT_RATE -> 0: none
void ak_synth_free (ak_synth *s);
void ak_synth_midi (ak_synth *s, int channel, int command, int data1, int data2); // 0x90 note on, 0x80 off, 0xC0 program...
void ak_synth_render (ak_synth *s, short *out, int frames); // never allocates, never calls the kernel
```

the FM synthesizer: the system's voices (fmsynth.h; they were the kernel's until 2026-10-05)

16 voices, each a plain wave or a two-operator FM instrument, played by the player's thread, mixed with the file and the MIDI notes. wave and the instrument are the kapi's names (SOUND_SQUARE ... SOUND_FM, struct kapi_fm_instrument). A note sounds until it is stopped.

```
struct kapi_fm_instrument;
int ak_fm_instrument (int voice, const struct kapi_fm_instrument *ins); // 0 / -1
int ak_fm_start (int voice, unsigned milli_hz, int wave, int volume); // voice 0..15, volume 0..255
void ak_fm_stop (int voice); // -1: all (the releases are heard)
void ak_fm_silence (void); // all silent at once
void ak_fm_render (short *out, int frames); // the voices' next frames written to out (off line: an export)
void ak_fm_live (int on); // 0: the player leaves the voices alone -- the program renders them (ak_fm_render); 1 (at first): heard
```

effects (FPU programs): MeltySynth's reverb (Freeverb) and chorus

```
typedef struct ak_reverb ak_reverb;
ak_reverb *ak_reverb_new (int rate);
void ak_reverb_set (ak_reverb *r, float room, float damp, float wet, float width); // < 0:
unchanged
void ak_reverb_process (ak_reverb *r, const float *in, float *left, float *right, int
frames); // mono in, stereo out
void ak_reverb_mute (ak_reverb *r);
void ak_reverb_free (ak_reverb *r);
typedef struct ak_chorus ak_chorus;
ak_chorus *ak_chorus_new (int rate, float delay_s, float depth_s, float hz); //
(MeltySynth's own: 0.002, 0.0019, 0.4)
void ak_chorus_process (ak_chorus *c, const float *inL, const float *inR, float *outL, float
*outR, int frames);
void ak_chorus_mute (ak_chorus *c);
void ak_chorus_free (ak_chorus *c);

}
#endif
```